

Ryan Cho

Full Stack Engineer | AI Systems

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PROFESSIONAL SUMMARY

Full stack engineer with 6+ years delivering production web platforms and AI-enabled product capabilities across fintech and enterprise environments. Experienced building end-to-end features with Next.js, Node.js, and Python, integrating LLMs, RAG pipelines, and external APIs into dependable workflows. Owns architecture through deployment on Docker and Kubernetes, with focus on streaming interactions, observability, and resilient recovery for multi-step agent experiences.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Led **end-to-end delivery** of AI-native product features across **Next.js** frontends and **Node.js** services, moving workflows beyond chat into persistent, goal-driven experiences used daily by investment teams.
- Designed **multi-step agent workflows** with planning, tool invocation, failure detection, and recovery paths, improving task completion reliability across sessions and external integrations.
- Integrated **OpenAI** and **Claude** models with memory stores and third-party APIs, establishing reusable abstractions that reduced duplicate integration work across product surfaces.
- Built **real-time streaming interfaces** for partial LLM responses under tight latency constraints, balancing responsiveness with output quality through progressive rendering and client-side state management.
- Implemented **RAG pipelines** in **Python** for document retrieval and contextual generation, pairing vector search with structured SQL lookups to ground answers in approved firm data.
- Architected **REST and event-driven APIs** supporting agent orchestration, tool routing, and audit logging, enabling cross-team feature delivery without tight coupling between services.
- Deployed services with **Docker** and **Kubernetes**, defining health checks, rollout strategies, and environment-specific configuration for stable production releases.
- Strengthened **observability and fallback mechanisms** with structured logging, tracing, and degraded-mode paths for LLM and tool failures, cutting mean time to recovery during peak usage.
- Partnered with product, ML, and backend stakeholders to **iterate from production failure modes**, using usage signals and evaluation runs to raise workflow success rates release over release.

Technology Analyst, Incoming Technology Analyst, Technology

May/2019 - Apr/2022

Summer Analyst,

BNP Paribas

Remote | Seventh Avenue, New York

- Delivered **full stack enhancements** for internal banking platforms, combining React-based UIs with **Node.js** and Java services supporting analysts across global markets operations.
- Designed **API contracts and service boundaries** for trade reporting and reference data workflows, improving maintainability as downstream teams adopted shared integration patterns.
- Built **Python** utilities and batch processors for data validation and reconciliation, reducing manual review effort on high-volume operational datasets.
- Optimized **SQL** queries and indexing strategies across PostgreSQL and Oracle stores, improving report generation performance for time-sensitive regulatory submissions.
- Introduced **containerized deployment** practices with Docker for internal tools, standardizing local development and reducing environment drift between QA and production.
- Partnered with business analysts and operations leads to translate ambiguous requirements into **pragmatic technical designs**, shipping incremental value under evolving compliance constraints.
- Implemented **monitoring dashboards and alerting** for critical batch jobs and API endpoints, giving on-call teams faster visibility into failures before they impacted downstream consumers.

- Mentored incoming analysts on **code review standards**, testing practices, and release procedures, helping new hires contribute production changes within their first sprint cycles.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Contributed to **full stack feature development** for a SaaS platform, implementing UI components and backend endpoints that supported customer onboarding and account management flows.
- Built **REST APIs** in Node.js with input validation and structured error responses, enabling the frontend team to ship interactive workflows with predictable integration behavior.
- Developed **React** views with reusable component patterns, improving consistency across admin and customer-facing screens while reducing duplicate UI logic.
- Integrated **third-party payment and notification services**, handling webhook processing and retry logic for transient provider failures in production-like staging environments.
- Participated in **code reviews and sprint planning**, gaining early ownership of small production fixes and learning team standards for testing, deployment, and incident follow-up.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Built and maintained **responsive marketing pages** for a nonprofit foundation, improving content layout and mobile readability for donor-facing information.
- Implemented **HTML, CSS, and JavaScript** updates across existing site templates, aligning visual presentation with updated brand guidelines from stakeholders.
- Configured **form submission handling** and basic analytics hooks, giving the team clearer visibility into visitor engagement during campaign periods.
- Collaborated with foundation staff to **prioritize content changes**, delivering iterative updates on a short timeline with lightweight version control practices.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: Next.js, React, TypeScript, JavaScript, HTML5, CSS3, Streaming UI, Component Architecture

Backend: Node.js, Python, REST APIs, Event-Driven Services, LLM Integration, Agent Orchestration, OpenAI, Claude

Databases: PostgreSQL, SQL, NoSQL, MongoDB, Vector Stores, Redis, Schema Design, Query Optimization

Cloud & DevOps: Docker, Kubernetes, CI/CD, AWS, Container Orchestration, Infrastructure as Code, Environment Management

Monitoring & Tools: Observability, Structured Logging, Distributed Tracing, Grafana, Prometheus, Git, Open Source Tooling

Workflow & PM: Agile, Cross-Functional Collaboration, Technical Design Docs, Code Review, Stakeholder Communication, Feature Ownership